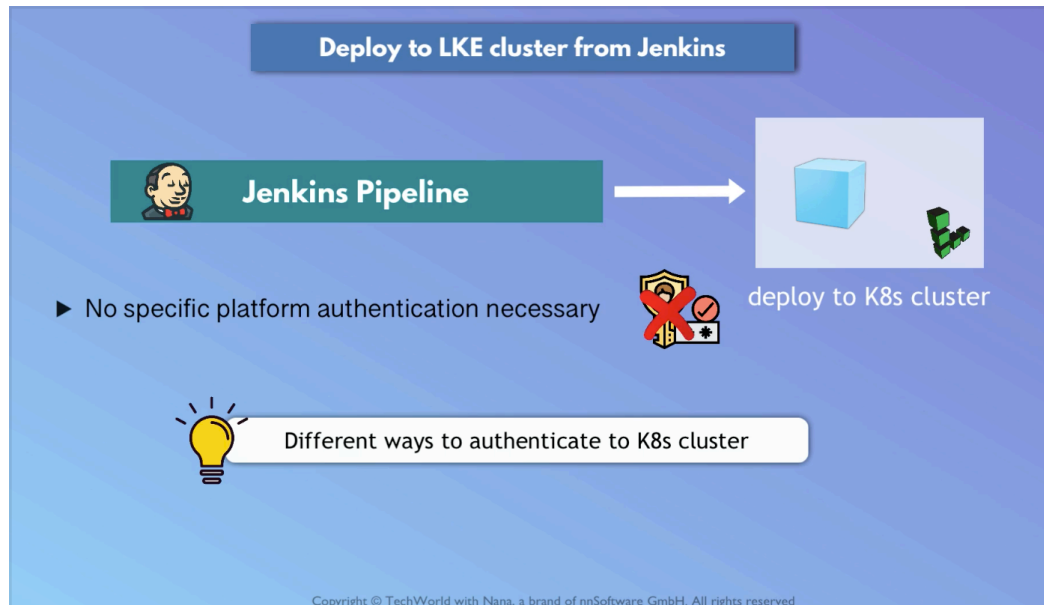


Module 11 - chapter 7

BONUS: Deploy to LKE Cluster from Jenkins Pipeline

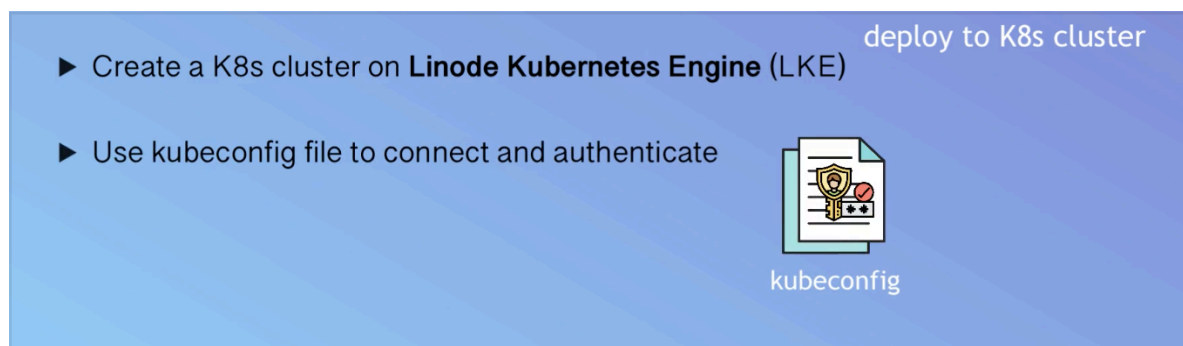
Jenkinsfile configuration is different depending on the cloud provider:



In AWS we use the below to authenticate:

```
20 stage('deploy') {
21     environment {
22         AWS_ACCESS_KEY_ID = credentials('jenkins_aws_access_key_id')
23         AWS_SECRET_ACCESS_KEY = credentials('jenkins_aws_secret_access_key')
24     }
```

For Linode, this is not the case:



Steps we need to configure:

- Install kubectl -> we already have this ✓
- Install Kubernetes CLI Jenkins Plugin -> we will add Linode kube config file as a credential so we need the plugin to load the kubeconfig file (in aws we created the config file manually, here we will load it) -> we will execute kubectl with kubeconfig credentials

- Configure Jenkins to deploy to LKE cluster using the plugin

Steps we need to configure

- ▶ **Kubectl** command line tool available inside Jenkins container
- ▶ **Install Kubernetes CLI Jenkins Plugin**
Execute kubectl with kubeconfig credentials
- ▶ **Configure Jenkinsfile** to deploy to LKE cluster

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Create k8s cluster on Linode and connect to it

Only One node for this cluster.

This is similar to bare metal clusters -> we will be connecting to the machines only we are not connecting in addition to a platform like in the case of AWS.

I download the kubeconfig file:

Summary

Version 1.35	1 CPU Core	Kubernetes API Endpoint:
GB, London 2	2 GB RAM	https://66e0eabb-2698-4b33-beff-3d70ba09c015.uk-lon-1-gw.linodek8s.net:443
\$12.00/month	50 GB Storage	Kubeconfig:
		test-kubeconfig.yaml View Copy Token Reset

Cluster ID: 615340 | Control Plane ACL: Enable | Created: 2026-06-10 08:22 | Updated: 2026-06-10 08:22

Recent download history

- test-kubeconfig (1).yaml
2,821 B • 5 minutes ago
- jenkins_accessKeys.csv
99 B • 19 hours ago

I created a folder called 'eks' and copy the file there:

File manager view showing the **eks** folder containing the file **test-kubeconfig.yaml**.

In my terminal I am currently connected to AWS EKS, which I deleted yesterday:
-> kubectl get pod

We can see there are no pods as I delete the whole cluster

```
astrid@192.168.1.112 /Users/astrid
% kubectl get pod
E0610 09:25:27.660829    98489 memcache.go:265] "Unhandled Error" err="couldn't get current server API group list: Get \"http://localhost:8080/a
pi?timeout=32s\": dial tcp 127.0.0.1:8080: connect: connection refused"
E0610 09:25:27.661781    98489 memcache.go:265] "Unhandled Error" err="couldn't get current server API group list: Get \"http://localhost:8080/a
pi?timeout=32s\": dial tcp 127.0.0.1:8080: connect: connection refused"
E0610 09:25:27.662609    98489 memcache.go:265] "Unhandled Error" err="couldn't get current server API group list: Get \"http://localhost:8080/a
pi?timeout=32s\": dial tcp 127.0.0.1:8080: connect: connection refused"
E0610 09:25:27.663474    98489 memcache.go:265] "Unhandled Error" err="couldn't get current server API group list: Get \"http://localhost:8080/a
pi?timeout=32s\": dial tcp 127.0.0.1:8080: connect: connection refused"
E0610 09:25:27.665159    98489 memcache.go:265] "Unhandled Error" err="couldn't get current server API group list: Get \"http://localhost:8080/a
pi?timeout=32s\": dial tcp 127.0.0.1:8080: connect: connection refused"
The connection to the server localhost:8080 was refused - did you specify the right host or port?
astrid@192.168.1.112 /Users/astrid
```

I need to connect to Linode cluster and for that I need to set the env var KUBECONFIG to the path of the downloaded kubeconfig file:

```
astrid@192.168.1.112 /Users/astrid
% cd eks
astrid@192.168.1.112 /Users/astrid/eks
% ls
test-kubeconfig.yaml
astrid@192.168.1.112 /Users/astrid/eks
% █
```

-> export KUBECONFIG=~/.eks/test-kubeconfig.yaml

Now I can connect to Linode cluster:

-> kubectl get node

```
astrid@192.168.1.112 /Users/astrid/eks
% export KUBECONFIG=~/.eks/test-kubeconfig.yaml
astrid@192.168.1.112 /Users/astrid/eks
% kubectl get node
NAME                                STATUS    ROLES    AGE    VERSION
lke615340-902432-166c9b470000    Ready    <none>    9m41s    v1.35.3
astrid@192.168.1.112 /Users/astrid/eks
% █
```

Next, we need Jenkins to have access to the kubeconfig like we just did 🙏.

Add LKE credentials on Jenkins

We will create the new credential in the pipeline created in Module 11 - chapter 6 and we select 'secret file'

Jenkins EKS-chapter-11 / Credentials / Folder / Global credentials (unrestrict...

New credentials

Kind

- ✓ Username with password
- AWS Credentials
- GitHub App
- SSH Username with private key
- Secret file**
- Secret text
- Certificate

☐ Treat username as secret ?

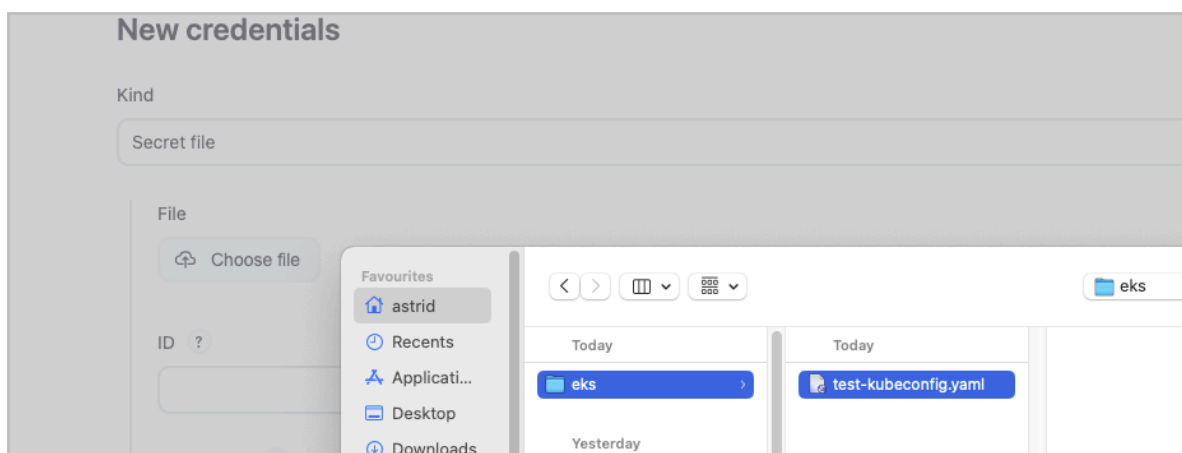
Password ?

ID ?

Description ?

Create

The load the kubeconfig file:



Then give an id:

Jenkins / EKS-chapter-11 / Credentials / Folder / Global credentials (unrestrict...

New credentials

Kind

Secret file

File

Choose file test-kubeconfig.yaml

ID ?

lke-credentials

Description ?

Create

And it is created:

Jenkins / EKS-chapter-11 / Credentials / Folder / Global credentials (unrestrict...

Global credentials (unrestricted)

Credentials that should be available irrespective of domain specification to requirements matching.

ID	Name
 jenkins_aws_access_key_id	jenkins_aws_access_key_id
 jenkins-aws_secret_access_key	jenkins-aws_secret_access_key
 lke-credentials	test-kubeconfig.yaml

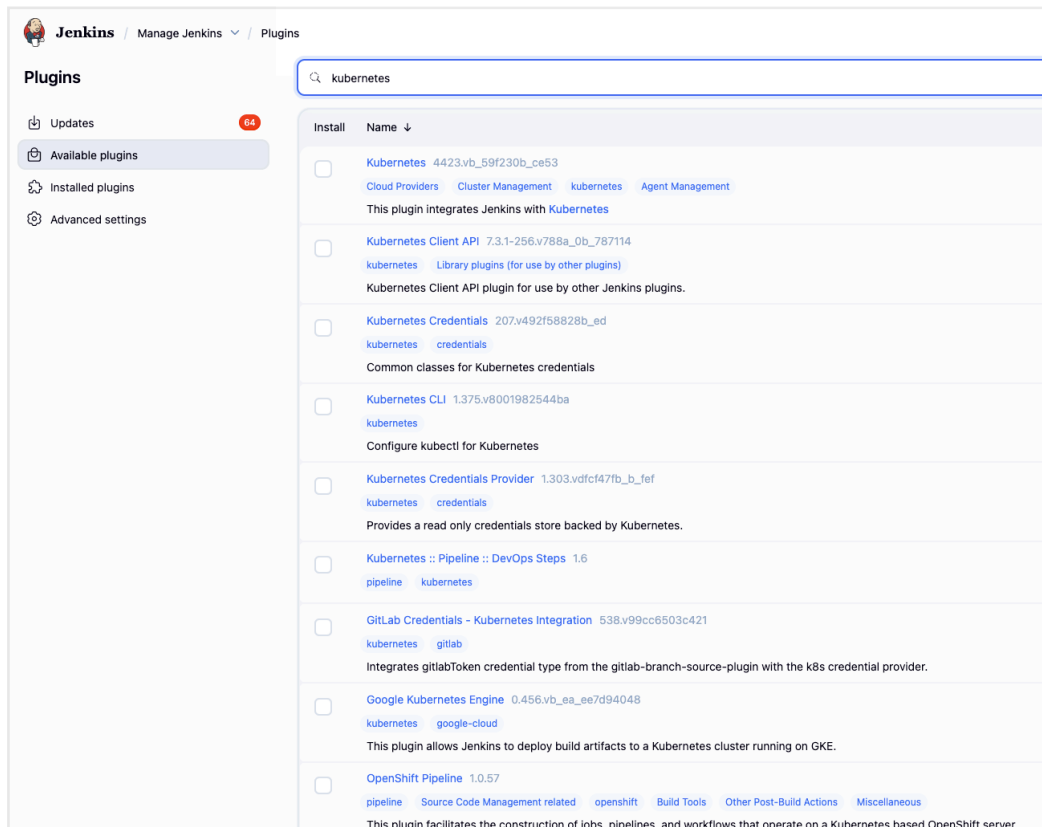
Icon: S M **L**

And it is a secret file:

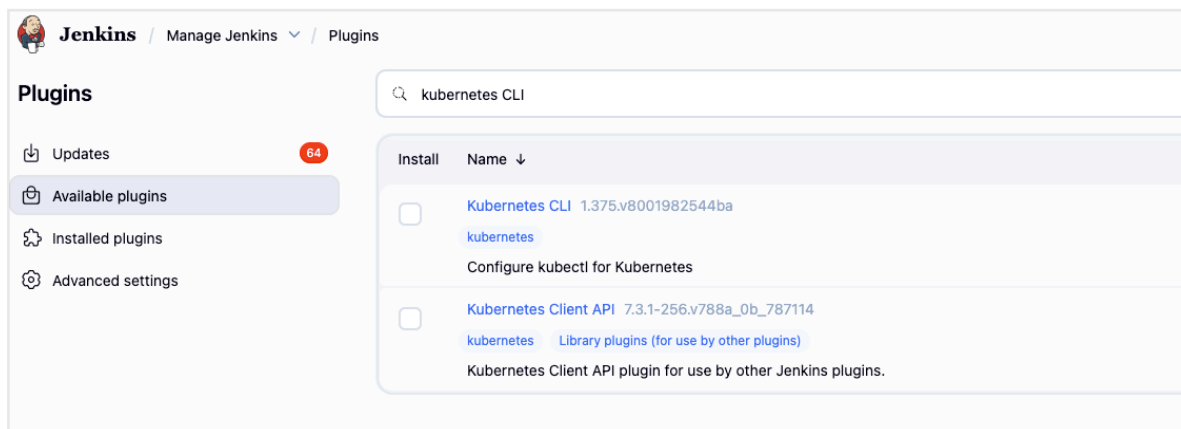
 lke-credentials	test-kubeconfig.yaml	Secret file	
Icon: S M L			

Install Kubernetes CLI Plugin on Jenkins

Jenkins has several Kubernetes plugins, the first one could be use but is more complex and it is preferred for more complex jenkins set ups like Jenkins clusters (multiple nodes for Jenkins jobs)



we will use a more simple plugin Kubernetes CLI

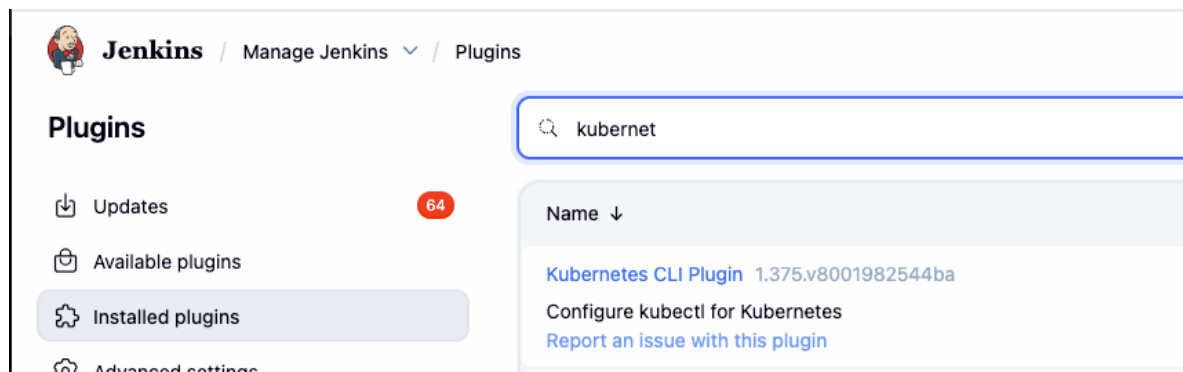


I installed it and restarted the container

-> docker start ec090f856ef8

```
root@jenkins-server:~# docker start ec090f856ef8
ec090f856ef8
root@jenkins-server:~# docker ps
CONTAINER ID   IMAGE                COMMAND                  CREATED        STATUS        PORTS
ec090f856ef8   jenkins/jenkins:lt  "/usr/bin/tini -- /u...  17 hours ago   Up 4 seconds   0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp
root@jenkins-server:~#
```

And the plugin is installed

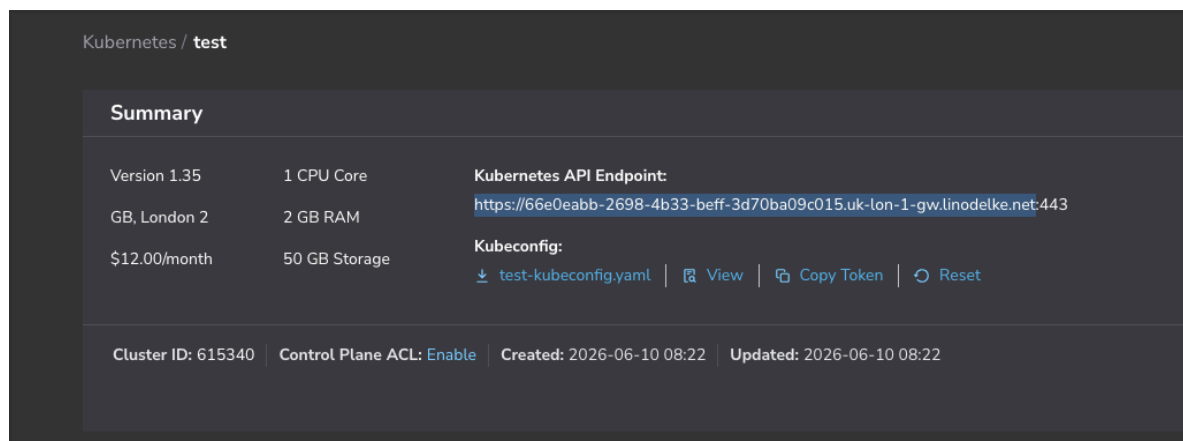


Configure Jenkinsfile to deploy to LKE cluster

We don't need environment code block that we used for AWS instead we use -> 'withKubeConfig'

and we pass:

- the Id of the credential we created for Linode in Jenkins -> 'lke-credentials'
- and we also pass the Linode server url (ignore :443 as it is the default)
-> https://66e0eabb-2698-4b33-beff-3d70ba09c015.uk-lon-1-gw.linodelke.net



I updated the pipeline config to only run branch 'Module_11/EKS_LKE_chapter-7':

Jenkins / EKS-chapter-11 / Configuration

Configuration

- General
- Branch Sources**
- Build Configuration
- Scan Repository Triggers
- Orphaned Item Strategy
- Appearance
- Health metrics
- Properties

Discover pull requests from origin ?

Strategy ?

The current pull request revision

Discover pull requests from forks ?

Strategy ?

The current pull request revision

Trust ?

From users with Admin or Write permission

May not be supported on older versions of GitHub Enterprise. See help button.

Filter by name (with regular expression)

Regular expression ?

Module_11/EKS_LKE_chapter-7

Add ▾

Property strategy

All branches get the same properties

I saved the pipeline config and run the job And it worked 🥳

Jenkins / EKS-chapter-11 / Module_11/EKS_LKE_chapter-7

Status

- Changes
- Build Now
- View Configuration
- Full Stage View
- Stages
- GitHub
- Pipeline Syntax
- Credentials

Buils

Filter

Today

#1 9:17 AM

Module_11/EKS_LKE_chapter-7

Full project name: EKS-chapter-11/Module_11%2FEKS_LKE_chapter-7

Stage View

	Declarative: Checkout SCM	build app	build image	deploy
Average stage times: (full run time: ~14s)	4s	355ms	194ms	2s
#1 Jun 10 10:17 No Changes	4s	355ms	194ms	2s

Permalinks

- Last build (#1), 59 sec ago
- Last stable build (#1), 59 sec ago
- Last successful build (#1), 59 sec ago
- Last completed build (#1), 59 sec ago

```
Jenkins / EKS-chapter-11 / Module_11/EKS_LKE_chapter-7 / #1 / Console Output

[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (build image)
[Pipeline] script
[Pipeline] {
[Pipeline] echo
building the docker image...
[Pipeline] }
[Pipeline] // script
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (deploy)
[Pipeline] script
[Pipeline] {
[Pipeline] echo
deploying docker image...
[Pipeline] withKubeConfig
[Pipeline] {
[Pipeline] sh
+ kubectl create deployment nginx-deployment --image=nginx
deployment.apps/nginx-deployment created
[Pipeline] }
[kubernetes-cli] kubectl configuration cleaned up
[Pipeline] // withKubeConfig
[Pipeline] }
[Pipeline] // script
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline

GitHub has been notified of this commit's build result

Finished: SUCCESS
```

And from my laptop I check the pod 🤖
-> kubectl get pod

```
astrid@192.168.1.112 /Users/astrid/eks
% kubectl get pod
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6ff797d4c9-j69vg  1/1     Running   0           3m33s
astrid@192.168.1.112 /Users/astrid/eks
% █
```

It was easier without the additional platform authentication ☆

Depends on the Platform you use



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